

About us:

We are the Brazil team Águabots, representing the Farias Brito educational institution. The name Águabots was chosen to symbolize the fluidity, adaptability, and intelligence of our robots, just like the importance of water for life and our mission to rescue victims in disaster situations, together with the following members:



Isaac Newton S. Araújo

- Responsible for AI and Software Architecture



Manuella A. N. Magalhães

- Responsible for mapping



Isabella F. G. P. Araujo

- Responsible for the design and path planning



Vinícius F.S. Oliveira

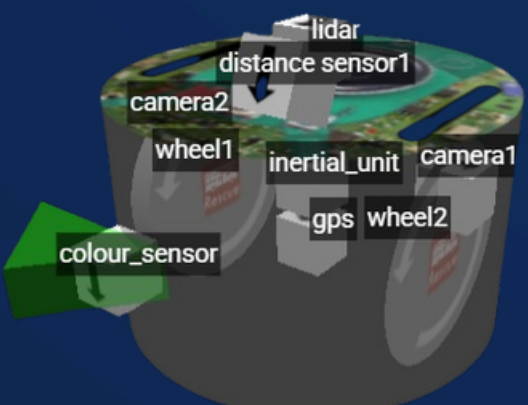
- Responsible for navigation and movement



Awards:

1st place in the Brazilian Robotics Competition (CBR) Rescue Simulation 2025.

HARDWARE



Source: Authors, 2026
ROBOT COMPONENTS



Source: Authors, 2026
ROBOT IN SIMULATION

Wheels: Two side-mounted wheels are responsible for the robot's movement, allowing it to move forward, make smooth turns, and rotate on its own axis.

Inertial Unit: Used to ensure that the robot makes turns with exact 90° precision.

Color Sensor: Used to identify the color of the simulated arena's floor. It allows the detection of holes that cannot be crossed.

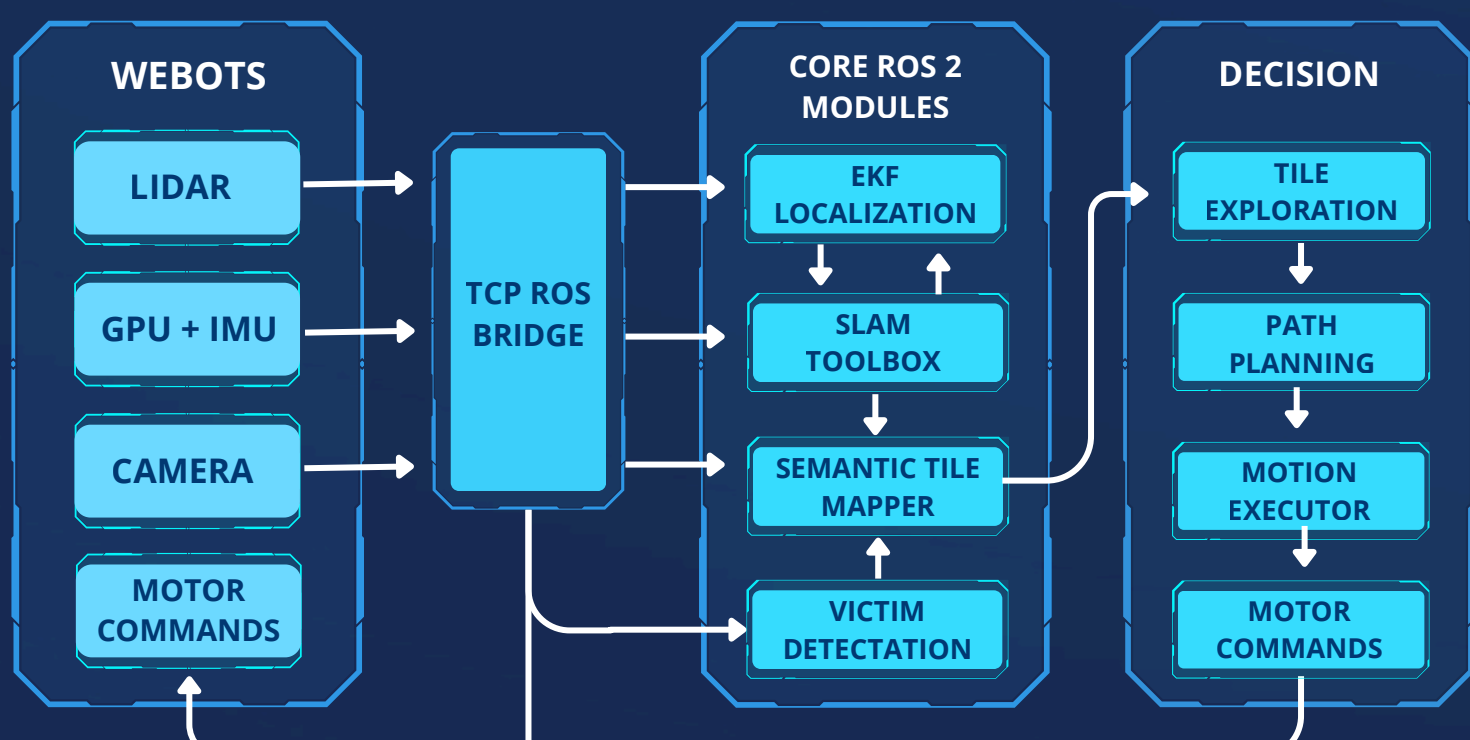
GPS: Used to record and transmit the precise location of victims detected by the robot. It is also used to assist in mapping.

LiDAR: Used for accurate wall detection in the environment. The data provided by the LiDAR is essential to avoid collisions and to build representations of the traversed space.

Cameras: Two cameras are positioned on the sides of the robot, assisting in precise victim detection through computer vision. Both cameras have a resolution of 64x64.

Distance Sensor: Used to detect holes. It compensates for the color sensor's limitation of analyzing only one point, making hole detection more reliable.

SYSTEM ARCHITECTURE



Flowchart: Authors, 2026

ÁGUABOTS

INCHEON- ROBOCUP 2026 - RESCUE SIMULATION



BRAZIL

NAVIGATION AND EXPLORATION

Exploration

Custom tile-based exploration combining semantic mapping, LiDAR clearance analysis, backtracking and failed-goal memory.

Goal Selection

Selects the next open unvisited neighbor tile or initiates backtracking when no new tile is reachable.

Planning

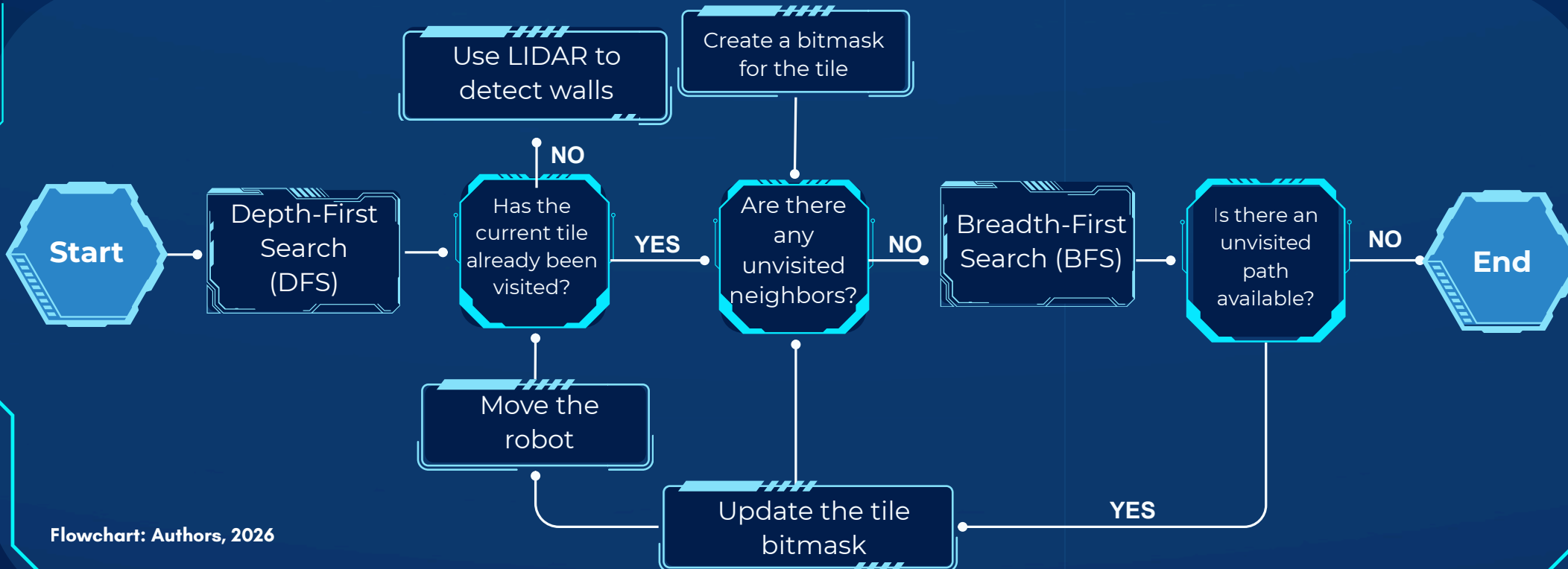
DFS-style planning with weighted backtracking through visited non-blocked tiles.

Motion Control

Converts tile movements into calibrated Webots wheel-position targets.

Safety & Recovery

Cancels unsafe movements, reverses partial motion, marks blocked tiles, and ensures return-to-start behavior.



Flowchart: Authors, 2026

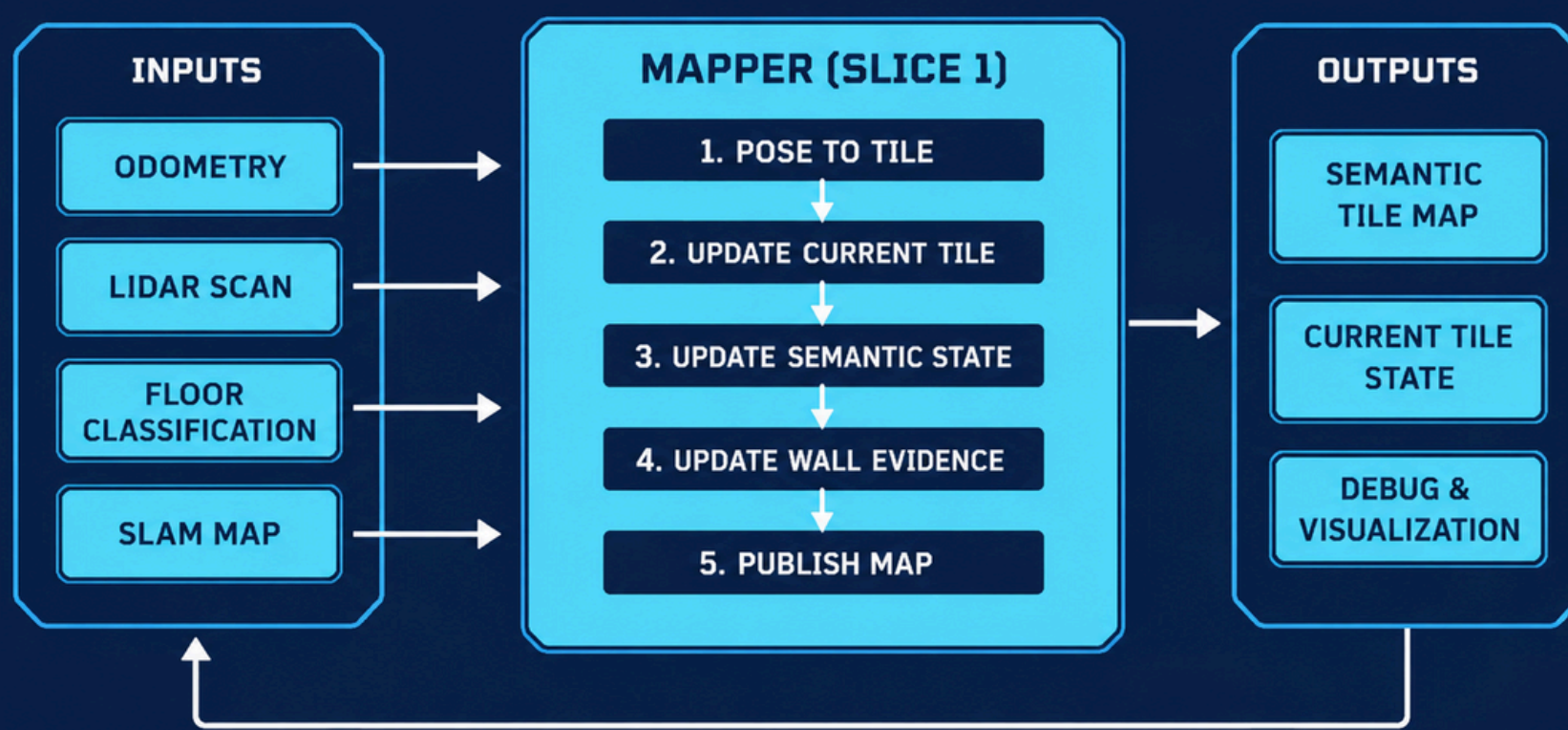
MAPPING

The mapping system represents the Rescue Simulation maze as a semantic tile-based map, since the arena is organized as a grid of square tiles

The mapper receives standardized ROS 2 data, mainly filtered odometry and LiDAR scans. Odometry provides the robot's position, while LiDAR detects nearby walls and obstacles. Ground sensors classify each tile as normal, colored, swamp, hole.

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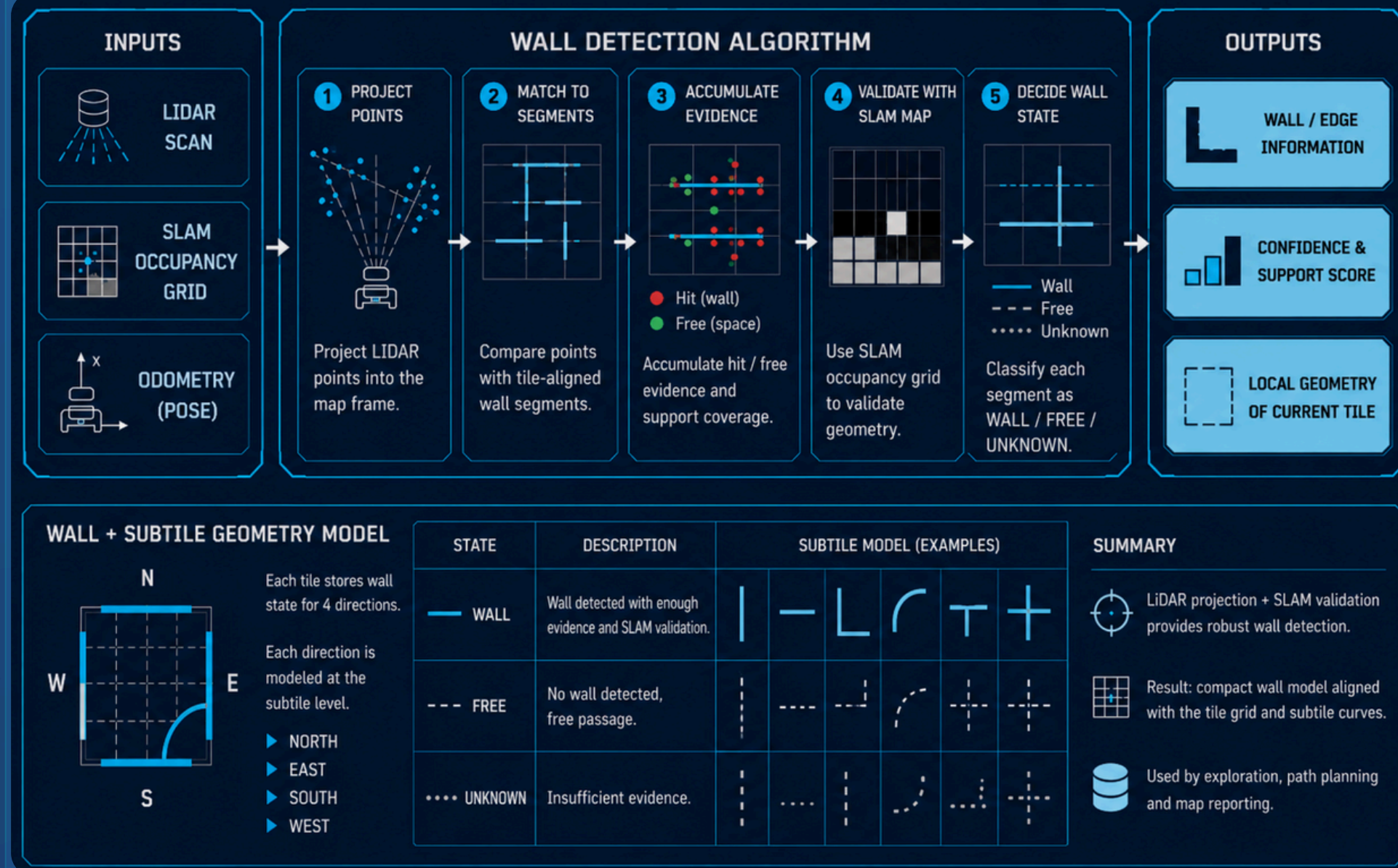
The mapper converts sensor observations into a semantic tile representation of the maze.



Source: Authors, 2026

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Wall and geometry mapping



Source: Authors, 2026

VICTIM RECOGNITION

Image Processing

- Frame filtering
- Duplicate detection removal

Object Detection

Lightweight ONNX CNN

- Detects: Target and victim

Target Recognition

CNN + Computer Vision

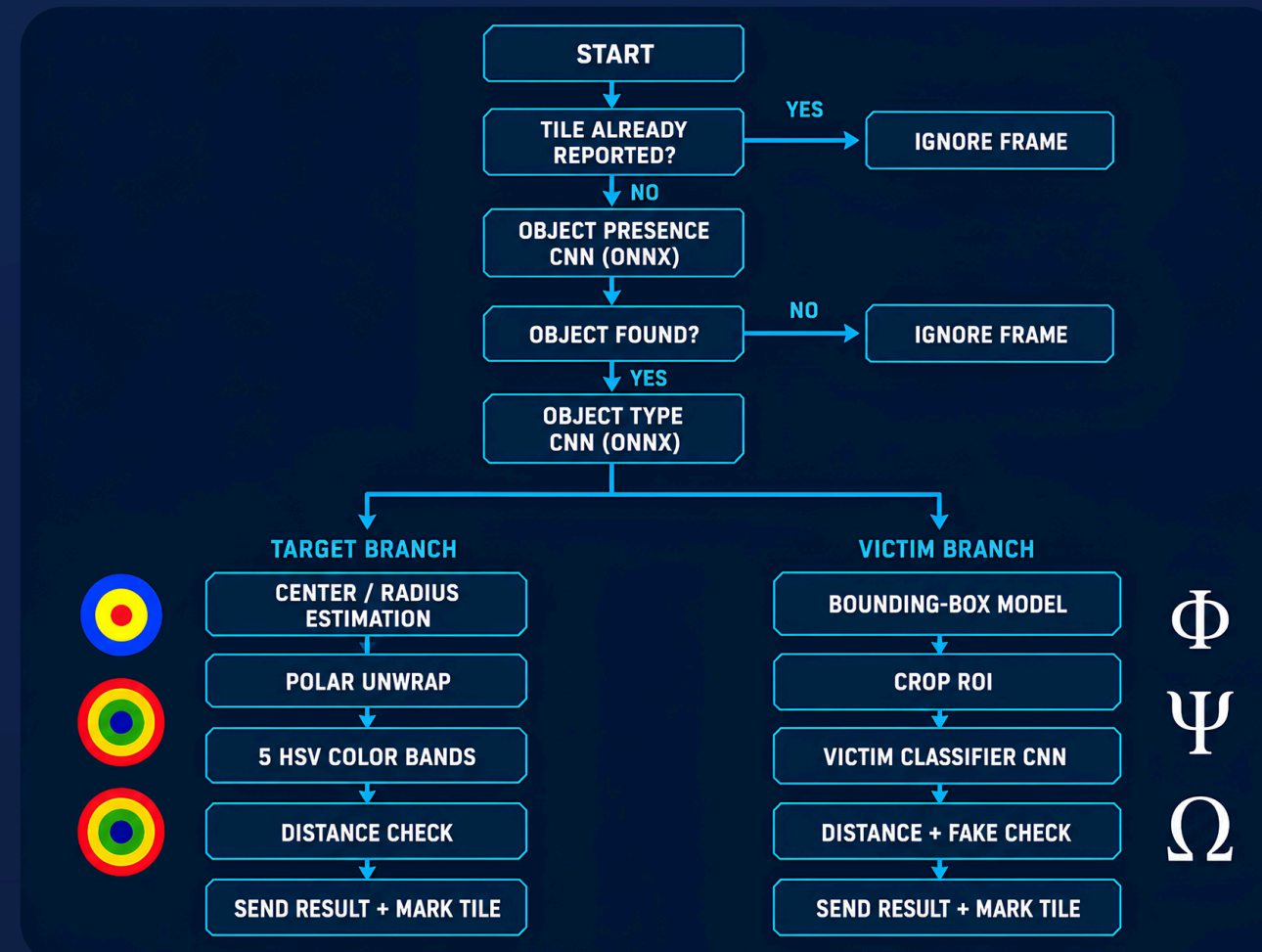
- Circle detection
- Polar unwrapping
- HSV color decoding
- Distance validation

Victim Recognition

CNN Pipeline

- Bounding box detection
- Image classification
- Fake victim rejection

Training & Deployment



Flowchart: Authors, 2026